“SHARED MENTAL MODEL-INSPIRED STRUCTURED MEMORY IN GENERATIVE AI-BASED MULTI-AGENT SYSTEM: A SIMULATION STUDY”

Guide to Construct Validity “Communication”

Communication quality concerns whether initially distributed, decision-relevant information is shared and subsequently taken up by other team members.

A fact counts as communicated when:

1. The fact was initially private to one agent.
2. That agent mentions it in the public discussion.
3. A different agent later mentions the same fact.
4. The later mention occurs chronologically after the initial mention.
5. Paraphrases count if they clearly express the same fact.
6. Each fact is counted at most once.
7. Public tool questions and answers are included.

The communication score is:

$$\text{Communication Quality} = \frac{\text{Number of communicated facts}}{24}$$

Guide to Construct Validity “Cooperation”

Cooperation quality concerns whether team members engage with, discuss, compare, and integrate one another’s information and assessments into a collective evaluation, corresponding to the concept of information elaboration.

A message counts as cooperative when:

1. The message is an ordinary agent message, not a tool call or tool-agent message.
2. The message is not the first message in the discussion.
3. The message explicitly engages with or builds on another agent’s contribution.
4. The message transforms candidate-related information by doing at least one of the following:
    a. comparing candidates or pieces of evidence;
    b. combining candidate-related evidence;
    c. drawing an inference about a candidate;
    d. qualifying or discussing a concern, limitation, or risk regarding a candidate;

e. revising, changing, or updating a recommendation, vote, or assessment.

5. A message is counted at most once.

The cooperation score is:

$$\text{Cooperation Quality} = \frac{\text{Number of cooperative eligible ordinary agent messages}}{\text{Total number of eligible ordinary agent messages}}$$

Private Facts

- **Agent 1:**
- Candidate A is sometimes not good at taking criticism.
- Candidate A can be unorganized.
- Candidate B can be grumpy.
- Candidate B can be uncooperative.
- Candidate C is known to be 100% reliable.
- Candidate C creates a positive atmosphere with his crew.
- Candidate D is regarded as arrogant.
- Candidate D has relatively weak leadership skills.
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- **Agent 2:**
- Candidate A is regarded as a show-off.
- Candidate A is regarded as being not open to new ideas.
- Candidate B has a relatively weak memory for numbers.
- Candidate B makes nasty remarks about his colleagues.
- Candidate C keeps calm in a crisis.
- Candidate C understands complicated technology.
- Candidate D is regarded as a know-it-all.
- Candidate D has a hot temper.
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- **Agent 3:**
- Candidate A is unfriendly.
- Candidate A eats unhealthily.
- Candidate B is regarded as pretentious.
- Candidate B sometimes adopts the wrong tone when communicating.
- Candidate C puts concern for others above everything.
- Candidate C has excellent attention skills.
- Candidate D is considered moody.
- Candidate D is regarded as a loner.

Public Facts:

- Candidate A can anticipate dangerous situations.
- Candidate A is able to see complex connections.
- Candidate A has excellent spatial vision.
- Candidate A has very good leadership qualities.
- Candidate B is very conscientious.
- Candidate B handles stress very well.
- Candidate B is good at assessing weather conditions.
- Candidate B has excellent computer skills.
- Candidate C can make correct decisions quickly.
- Candidate C has difficulty communicating ideas.
- Candidate C is regarded as egocentric.
- Candidate C is not very willing to further his education.
- Candidate D responds to unexpected events adequately.
- Candidate D can concentrate very well.
- Candidate D solves problems extremely well.
- Candidate D takes responsibility seriously.